

Polarization states	Cytokine drivers	Top marker genes
B-a	IFNα1, IFNβ, IFNε, IFNκ	*Ifi27l2a, Ifit3, Trim30a, Slfn5, Shisa5*
B-b	IFNγ, IL-2, IL-12, IL-15, IL-18	*Serpina3g, Stat1, Ifi47, Socs1, Gbp7*
B-e	CD40L, IL-21	*Ncl, Eif4a1, Nme1, Eif5a, Ptma*
B-f	IL-4	*St7, Samsn1, Tmem123, Il4i1, Dock10*
T4-a	IFNα1, IFNβ, IFNε, IFNκ	*Ifi27l2a, Isg15, Ifit1, Ifit3, Slfn5*
T4-b	IFNγ, IL-2, IL-12, IL-15, IL-18	*Igtp, Stat1, Ifi47, Gbp2, Iigp1*
T4-c	IL-1α, IL-1β	*Igfbp4, Ndrg3, Bcl3, Ly6a, Mxd1, Ifngr1*
T4-e	IL-36α, TL1A	*Ptma, Eif5a, Cd82, Ncl, Set*
T4-f	IL-4, IL-7, TSLP	*Bcl2, Srm, Lta, Socs1, Rexo2*
T8-a	IFNα1, IFNβ, IFNε, IFNκ	*Isg15, Ifit3, Ifit1, Bst2, Usp18*
T8-b	IFNγ, IL-2, IL-12, IL-15, IL-18	*Stat1, Ifi27l2a, Igtp, Psmb9, Gbp4*
T8-c	IL-1α, IL-1β	*Igfbp4, Gadd45g, Bcl3, Ly6a, Tcf7, Ifngr1*
T8-e	IL-2, IL-15, IL-36α	*Ncl, Ptma, Eif5a, Npm1, Ran*
T8-f	IL-4, IL-7	*Eif5a, Ptma, Eif4a1, Ncl, Nme1*
Tgd-a	IFNα1, IFNβ, IFNε, IFNκ	*Isg15, Ifit1, Bst2, Ifi203, Rnf213*
Tgd-b	IFNγ, IL-2, IL-12, IL-15, IL-18	*Igtp, Stat1, Gbp4, Ifi47, Gbp2*
Tgd-c	IL-1α, IL-1β	*Npm1, Ncl, Eif5a, Gzmc, Odc1*
Tgd-d	TNF, TL1A, IL-17E	*Odc1, Adam8, Ltb, Cd82, Cyba*
Tgd-e	IL-2, IL-7, IL-15, IL-18, IL-36α	*Eif5a, Ppa1, Hspa5, Igtp, Ifi47*
Tgd-f	IL-23	*Cpd, Il22, Gzmb, Serpinb1a, Ffar2*
Treg-a	IFNα1, IFNβ, IFNε, IFNκ	*Ifit1, Isg15, Ifit3, Ifi27l2a, Bst2*
Treg-b	IFNγ, IL-2, IL-12, IL-15, IL-18	*Stat1, Iigp1, Gbp7, Igtp, Ifi47*
Treg-c	IL-1α, IL-1β, IL-36α, TNF	*Crem, Tnfrsf9, Tnfrsf18, Odc1, Tnfrsf4*
Treg-d	IL-18, IL-33, TL1A, TNF	*Ncl, Ptma, Mif, Eif5a, Npm1*
NK-a	IFNα1, IFNβ, IFNε, IFNκ	*Isg15, Bst2, Trim30a, Ms4a4b, Zbp1*
NK-c	IL-1α, IL-1β	*Ifngr1, Crem, Vps37b, Emb, Sh2d2a*
NK-e	IL-2, IL-12, IL-15	*Gzmb, Npm1, Ncl, Nme1, Eif5a*
NK-f	IL-18	*Myc, Xcl1, Traf1, Gadd45b, Cd83, Batf*
pDC-a	IFNα1, IFNβ, IFNε, IFNκ	*Isg15, Irf7, Phf11b, Ifi203, Slfn5*
pDC-b	IFNγ, IL-2, IL-15, IL-18	*Iigp1, Igtp, Gbp4, Stat1, Csf2rb2*
pDC-c	IL-1α, IL-1β	*Fkbp5, Zfp36l2, Klf13, Ptprc, Gpr171*
pDC-d	TNF, IL-3, GM-CSF	*Ly6d, Sub1, Fdps, Lcp1, Napsa*
pDC-e	IL-36α	*Bcl2a1d, Bcl2a1b, Il4i1, Tmem176a, Wfdc21*
cDC1-a	IFNα1, IFNβ, IFNε, IFNκ	*Irf7, Isg15, Slfn5, Ifi204, Rnf213*
cDC1-b	IFNγ, IL-2, IL-15, IL-18	*Cxcl9, Serpina3g, Zyx, Samhd1, Iigp1*
cDC1-c	IL-1α, IL-1β	*Wfdc17, Ccl17, Fth1, Mkrn1, Kmo*
cDC1-d	TNF	*Gadd45b, Etv3, Btg1, Ccr7, Basp1*
cDC1-e	GM-CSF, IL-3, TL1A	*Itgae, Prkcd, Ccl17, Cish, Cst3*
cDC1-f	IL-10	*Ifitm2, Wfdc17, Il1b, Gatm, Baz1a*
cDC2-a	IFNα1, IFNβ, IFNε, IFNκ	*Ifitm3, Isg15, Irf7, Phf11b, Rtp4*
cDC2-b	IFNγ, IL-2, IL-12, IL-15, IL-18	*Cxcl9, Gbp2, Ifi47, Gbp4, Samhd1*
cDC2-c	IL-1α, IL-1β, TNF	*Fabp5, Ccl17, Cish, Vdr, Socs2*
cDC2-e	IL-3, IL-33, LIF, OSM, CT-1	*Cyp7b1, Cd209d, Tcf4, Apod, Cd209e*
MigDC-a	IFNα1, IFNβ	*Isg15, Ifit1, Rnf213, Rsad2, Slfn5*
MigDC-c	IL-1α, IL-1β	*Cytip, Ccr7, Pkib, Slc27a3, Epb41*
MigDC-d	TNF	*Cd40, Ebi3, Serpinb9, Ccl22, Cd82*
MigDC-e	GM-CSF, IL-18	*Cish, Jaml, Nr4a3, Pkib, Coro2a*
LC-a	IFNα1, IFNβ	*Ifit1, Isg15, Ifit3, Oasl2, Oas3*
LC-b	IFNγ, IL-15	*Stat1, Cxcl9, Igtp, Gbp5, Irgm1*
LC-c	IL-1α, IL-1β	*Cytip, Ccr7, Cst3, Pik3r1, Tcaf2*
LC-d	TNF	*Cd40, Serpinb9, Srgn, Cd82, Ccl22*
LC-e	GM-CSF, IL-3, IL-18, TSLP	*Plek2, Gpbp1, Pkib, Nr4a3, Tcaf2*
Mac-a	IFNα1, IFNβ, IFNε, IFNκ	*Ifit3, Isg15, Ifit2, Rsad2, Irf7*
Mac-b	IFNγ, IL-2, IL-12, IL-15, IL-18	*Iigp1, Serpina3g, Cxcl9, Cxcl10, Serpina3f*
Mac-c	IL-1α, IL-1β, IL-36α	*Ccl9, Ccl6, Wfdc17, Ccl7, Sdc4*
Mac-d	TNF	*Saa3, Clec4e, Cxcl13, Il1b, Sod2*
Mac-e	IL-4, IL-13	*Chchd10, Glrx, Retnla, Ncl, Prps1*
Mono-a	IFNα1, IFNβ	*Ifit3, Rsad2, Ifit2, Irf7, Isg15*
Mono-b	IFNγ, IL-2, IL-12, IL-15, IL-18	*Cxcl10, Socs1, Gbp5, Fam26f, Ifi47*
Mono-c	IL-1α, IL-1β, IL-36α	*Sdc4, Srgn, Dab2, C5ar1, Adam8*
Mono-d	TNF	*Mmp14, Gpr84, Bcl2a1b, Nr1h3, Bcl2a1a*
Mono-e	GM-CSF, IL-3, IL-33	*Fabp5, Naaa, Eif5a, Cish, Dok2*
Neu-a	IFNα1, IFNβ, IFNγ	*Gbp2, Ifi47, Herc6, Gbp5, Isg15*
Neu-c	IL-1α, IL-1β	*Cxcl3, Cxcl2, Cish, Ier3, Srgn*
Neu-d	TNF	*Sod2, Ikbke, Marcksl1, Mapkapk2, Ccl3*
Neu-e	GM-CSF	*Hbegf, Sdad1, Ffar2, Car4, F2rl2*